PHYDSC401T — Mathematical Physics-III

Unit 1: Linear Vector Space and Matrices

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC401T, Full Marks 100 [ESE 70 + CCA 30] | Unit 1: 12 lectures

Notes 4 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Vectors in n-dimension, linear independence, basis and dimension.
2. Scalar product, norm and orthogonality, Schwarz inequality, Gram-Schmidt orthogonalization technique.
3. Linear operators and their matrix representation.
4. Eigenvalues and eigenvectors of a matrix, Cayley-Hamilton theorem.
5. Orthogonal, unitary and Hermitian matrices.
6. Infinite dimensional space, Hilbert space.

Top 2 high-mark topics in this unit

1. Linear independence, basis, dimension and rank of a matrix — 6 of the 12 AU questions in this unit are on LI/LD and rank, including 5-mark questions on rank by normal form and Gaussian elimination.
2. Eigenvalues, eigenvectors and Cayley-Hamilton theorem — the main derivation topic in the matrices half of the unit and a standard long-answer topic.

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1. Linear Vector Space, Linear Independence, Basis and Dimension

Vectors in n-Dimension and Vector Space Definition

An n-dimensional vector is an ordered n-tuple of real or complex numbers written as

$$v = (x_1, x_2, \ldots, x_n)$$

where each component x_i belongs to a scalar field F (typically $\mathbb{R}$ or $\mathbb{C}$).

A linear vector space V over a scalar field F is a set of elements called vectors, together with two operations: vector addition ($+ : V \times V \to V$) and scalar multiplication ($\cdot : F \times V \to V$). These operations must satisfy the following fundamental properties for all vectors $u, v, w \in V$ and all scalars $a, b \in F$:

The set V is closed under vector addition, such that

$$u + v \in V$$

Vector addition is commutative:

$$u + v = v + u$$

Vector addition is associative:

$$(u + v) + w = u + (v + w)$$

There exists a unique zero vector $0 \in V$ such that for any $v \in V$:

$$v + 0 = v$$

For every vector $v \in V$, there exists an additive inverse $-v \in V$ such that:

$$v + (-v) = 0$$

The set V is closed under scalar multiplication, meaning:

$$av \in V$$

Scalar multiplication is distributive over vector addition:

$$a(u + v) = au + av$$

Scalar multiplication is distributive over scalar addition:

$$(a + b)v = av + bv$$

Scalar multiplication is associative with respect to scalar multiplication:

$$a(bv) = (ab)v$$

The multiplicative identity scalar $1 \in F$ acts trivially on any vector:

$$1v = v$$

Linear Dependence and Linear Independence

Consider a set of k vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V. A linear combination of these vectors with scalar coefficients $\{c_1, c_2, \ldots, c_k\} \subset F$ set equal to the zero vector is written as:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0$$

The set of vectors $\{v_1, v_2, \ldots, v_k\}$ is defined to be linearly independent if the vector equation above holds if and only if all coefficients are identically zero:

$$c_1 = c_2 = \cdots = c_k = 0$$

If there exists at least one non-zero scalar coefficient $c_i \neq 0$ such that the linear combination sums to the zero vector, the set of vectors $\{v_1, v_2, \ldots, v_k\}$ is defined to be linearly dependent. In this case, at least one vector in the set can be expressed as a linear combination of the remaining vectors.

Tests for Linear Independence and Linear Dependence

For a singleton set containing a single vector $\{v\}$: Consider the relation

$$cv = 0$$

If $v \neq 0$, then $cv = 0$ forces $c = 0$. Hence, any singleton set containing a single non-zero vector is always linearly independent. Conversely, if $v = 0$, then $c \cdot 0 = 0$ holds for any non-zero scalar $c \neq 0$, making the set $\{0\}$ linearly dependent.

For a set of three vectors $\{v_1, v_2, v_3\}$ in $\mathbb{R}^3$: Let the vectors be represented in column form as $v_1 = (a_1, b_1, c_1)^T$, $v_2 = (a_2, b_2, c_2)^T$, and $v_3 = (a_3, b_3, c_3)^T$. We form a 3×3 matrix A using these vectors as columns:

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

The vector equation $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ is equivalent to the matrix equation:

$$A \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

If the determinant of A is non-zero,

$$\det(A) \neq 0$$

the matrix A is invertible, implying the only solution is $c_1 = c_2 = c_3 = 0$. Thus, the vectors are linearly independent if and only if $\det(A) \neq 0$. If $\det(A) = 0$, the system has non-trivial solutions, and the set of vectors is linearly dependent. Furthermore, any set containing more than n vectors in an n-dimensional space $\mathbb{R}^n$ (such as 4 vectors in $\mathbb{R}^3$) is automatically linearly dependent.

Basis and Dimension of a Vector Space

A set of vectors $B = \{e_1, e_2, \ldots, e_n\}$ in a vector space V forms a basis of V if it satisfies two conditions:

1. The set B is linearly independent.
2. The set B spans V, meaning every vector $v \in V$ can be written as a linear combination of the basis vectors:

$$v = \sum_{i=1}^{n} a_i e_i$$

where $a_i \in F$ are unique scalars called the components of v relative to the basis B.

The dimension of a vector space V, denoted $\dim(V)$, is defined as the number of vectors in any basis of V. If a basis consists of n vectors, then $\dim(V) = n$.

Rank of a Matrix

For an $m \times n$ matrix A: The row vectors of A span a subspace of F^n called the row space. The row rank of A is the maximum number of linearly independent row vectors in A.

The column vectors of A span a subspace of F^m called the column space. The column rank of A is the maximum number of linearly independent column vectors in A.

A fundamental theorem of linear algebra states that for any matrix A, the row rank equals the column rank. This common value is defined as the rank of the matrix, denoted $\text{rank}(A)$:

$$\text{row rank}(A) = \text{column rank}(A) = \text{rank}(A)$$

Since transposing a matrix A converts its rows into columns and its columns into rows, the row rank of A^T equals the column rank of A. Consequently:

$$\text{rank}(A^T) = \text{rank}(A)$$

The rank of a matrix can be evaluated by reducing it to its normal (canonical) form using elementary row and column operations. Elementary operations include interchanging two rows (or columns), multiplying a row (or column) by a non-zero scalar, and adding a scalar multiple of one row (or column) to another. Every $m \times n$ matrix A of rank r can be reduced to the canonical block matrix form:

$$N = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

where I_r is the $r \times r$ identity matrix, and the remaining blocks consist of zeros. The order r of the identity matrix I_r gives the rank of matrix A.

Row Reduction and Gaussian Elimination for Linear Systems

A system of m linear equations in n variables can be written in matrix form as $Ax = b$. Gaussian elimination involves forming the augmented matrix $[A|b]$ and applying elementary row operations to reduce A to row-echelon form, followed by back-substitution.

Consider the system of equations:

$$x + y + z = 6$$

$$2x - y + 2z = 6$$

$$2x + 2y + z = 9$$

We construct the augmented matrix $[A|b]$ representing the system:

$$[A|b] = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 2 & 6 \\ 2 & 2 & 1 & 9 \end{array}\right)$$

To eliminate x from the second row, we subtract twice the first row from the second row ($R_2 \to R_2 - 2R_1$):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2-2(1) & -1-2(1) & 2-2(1) & 6-2(6) \\ 2 & 2 & 1 & 9 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 2 & 2 & 1 & 9 \end{array}\right)$$

To eliminate x from the third row, we subtract twice the first row from the third row ($R_3 \to R_3 - 2R_1$):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 2-2(1) & 2-2(1) & 1-2(1) & 9-2(6) \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 0 & 0 & -1 & -3 \end{array}\right)$$

We divide the second row by -3 ($R_2 \to -\frac{1}{3}R_2$):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & -3 \end{array}\right)$$

We multiply the third row by -1 ($R_3 \to -R_3$):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

The matrix is now in upper-triangular row-echelon form. Writing out the corresponding system of equations gives:

$$x + y + z = 6$$

$$y = 2$$

$$z = 3$$

From the third row, we directly obtain:

$$z = 3$$

From the second row, we directly obtain:

$$y = 2$$

Substituting $y = 2$ and $z = 3$ into the equation from the first row yields:

$$x + 2 + 3 = 6$$

$$x + 5 = 6$$

$$x = 1$$

The unique solution to the system of linear equations is $x = 1$, $y = 2$, and $z = 3$.

2. Scalar Product, Norm, Orthogonality and the Schwarz Inequality

Scalar Product (Inner Product)

An inner product (or scalar product) on a vector space V over the field F (where F is $\mathbb{R}$ or $\mathbb{C}$) is a function mapping a pair of vectors $(u, v) \in V \times V$ to a scalar $\langle u, v \rangle \in F$, satisfying the following axioms for all $u, v, w \in V$ and $a, b \in F$:

The scalar product is positive-definite:

$$\langle v, v \rangle \geq 0$$

$$\langle v, v \rangle = 0 \iff v = 0$$

It satisfies conjugate symmetry:

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

For a real vector space, this reduces to symmetry:

$$\langle u, v \rangle = \langle v, u \rangle$$

It is linear in the first argument:

$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$$

From conjugate symmetry and linearity in the first argument, the scalar product is conjugate-linear in the second argument:

$$\langle u, av + bw \rangle = \bar{a}\langle u, v \rangle + \bar{b}\langle u, w \rangle$$

In the standard real Euclidean space $\mathbb{R}^n$, the scalar product of two vectors $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ is defined as:

$$\langle u, v \rangle = \sum_{i=1}^{n} u_i v_i$$

Norm of a Vector

The norm (or length) of a vector $v \in V$ induced by the scalar product is defined as the positive square root of the scalar product of the vector with itself:

$$\|v\| = \sqrt{\langle v, v \rangle}$$

The norm satisfies the following standard properties:

$$\|v\| \geq 0 \quad \text{and} \quad \|v\| = 0 \iff v = 0$$

$$\|cv\| = |c|\|v\| \quad \text{for any scalar } c \in F$$

$$\|u + v\| \leq \|u\| + \|v\| \quad \text{(Triangle Inequality)}$$

Orthogonality

Two vectors u and v in an inner product space V are said to be orthogonal if their scalar product is zero:

$$\langle u, v \rangle = 0$$

A set of vectors $\{e_1, e_2, \ldots, e_n\}$ is defined as an orthogonal set if every pair of distinct vectors in the set is orthogonal:

$$\langle e_i, e_j \rangle = 0 \quad \text{for all } i \neq j$$

An orthogonal set is called an orthonormal set if, in addition, every vector in the set has unit norm:

$$\langle e_i, e_j \rangle = \delta_{ij}$$

where δ_{ij} is the Kronecker delta, defined as $\delta_{ij} = 1$ when $i = j$ and $\delta_{ij} = 0$ when $i \neq j$.

Schwarz (Cauchy-Schwarz) Inequality

For any vectors u, v in an inner product space V, the absolute value of their scalar product is bounded by the product of their norms:

$$|\langle u, v\rangle| \leq \|u\|\|v\|$$

Proof of the Schwarz Inequality

If $v = 0$, then $\langle u, 0\rangle = 0$ and $\|u\|\|0\| = 0$. Thus, both sides of the inequality equal zero, and the equality holds trivially.

Now consider the case where $v \neq 0$. For any real scalar parameter $\lambda \in \mathbb{R}$, we construct the vector $w = u - \lambda v$. By the positive-definiteness axiom of the scalar product, the norm squared of w must be non-negative:

$$\langle u - \lambda v, u - \lambda v\rangle \geq 0$$

Expanding this expression using the linearity and symmetry properties of the scalar product yields:

$$\langle u, u - \lambda v\rangle - \lambda\langle v, u - \lambda v\rangle \geq 0$$

Applying symmetry to both terms gives:

$$\langle u, u\rangle - \lambda\langle u, v\rangle - \lambda\langle v, u\rangle + \lambda^2\langle v, v\rangle \geq 0$$

Using the definition of norm $\|u\|^2 = \langle u, u\rangle$ and $\|v\|^2 = \langle v, v\rangle$, along with the real symmetry relation $\langle v, u\rangle = \langle u, v\rangle$, we collect terms:

$$\|u\|^2 - 2\lambda\langle u, v\rangle + \lambda^2\|v\|^2 \geq 0$$

This expression is a quadratic polynomial in λ of the form $A\lambda^2 + B\lambda + C \geq 0$, where $A = \|v\|^2$, $B = -2\langle u, v\rangle$, and $C = \|u\|^2$. Since this quadratic in λ is non-negative for all real values of λ, we choose the specific value of λ that minimizes the expression. The minimum occurs at:

$$\lambda = \frac{\langle u, v\rangle}{\|v\|^2}$$

Substituting this specific choice of λ back into the quadratic inequality gives:

$$\|u\|^2 - 2\left(\frac{\langle u, v\rangle}{\|v\|^2}\right)\langle u, v\rangle + \left(\frac{\langle u, v\rangle}{\|v\|^2}\right)^2\|v\|^2 \geq 0$$

Multiplying out the second and third terms yields:

$$\|u\|^2 - \frac{2\langle u, v\rangle^2}{\|v\|^2} + \frac{\langle u, v\rangle^2\|v\|^2}{\|v\|^4} \geq 0$$

Simplifying the fraction in the third term gives:

$$\|u\|^2 - \frac{2\langle u, v\rangle^2}{\|v\|^2} + \frac{\langle u, v\rangle^2}{\|v\|^2} \geq 0$$

Combining the two like terms gives:

$$\|u\|^2 - \frac{\langle u, v\rangle^2}{\|v\|^2} \geq 0$$

Since $v \neq 0$, the norm $\|v\|^2$ is strictly positive. Multiplying the entire inequality by $\|v\|^2$ preserves the inequality sign:

$$\|u\|^2\|v\|^2 - \langle u, v\rangle^2 \geq 0$$

Rearranging the terms yields:

$$\langle u, v\rangle^2 \leq \|u\|^2\|v\|^2$$

Taking the non-negative square root on both sides of the inequality gives:

$$|\langle u, v\rangle| \leq \|u\|\|v\|$$

which completes the proof of the Schwarz inequality.

3. Gram-Schmidt Orthogonalization Technique

The Orthogonalization Algorithm

Given a set of linearly independent vectors $\{v_1, v_2, \ldots, v_n\}$ in an inner product space V, the Gram-Schmidt orthogonalization process constructs an orthogonal basis $\{u_1, u_2, \ldots, u_n\}$ and an orthonormal basis $\{e_1, e_2, \ldots, e_n\}$ spanning the same subspace.

The first orthogonal vector u_1 is chosen directly as the first input vector:

$$u_1 = v_1$$

The first orthonormal vector e_1 is obtained by normalizing u_1:

$$e_1 = \frac{u_1}{\|u_1\|}$$

The second orthogonal vector u_2 is obtained by subtracting from v_2 its orthogonal projection onto u_1:

$$u_2 = v_2 - \frac{\langle u_1, v_2\rangle}{\|u_1\|^2}u_1 = v_2 - \langle e_1, v_2\rangle e_1$$

The second orthonormal vector e_2 is obtained by normalizing u_2:

$$e_2 = \frac{u_2}{\|u_2\|}$$

For the k-th vector in general, u_k is constructed by subtracting from v_k its orthogonal projections onto all previously determined orthogonal vectors $u_1, u_2, \ldots, u_{k-1}$:

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle u_j, v_k \rangle}{\|u_j\|^2} u_j = v_k - \sum_{j=1}^{k-1} \langle e_j, v_k \rangle e_j$$

The k-th orthonormal vector e_k is obtained by normalizing u_k:

$$e_k = \frac{u_k}{\|u_k\|}$$

Repeating this process for all $k = 1, 2, \ldots, n$ yields an orthonormal set $\{e_1, e_2, \ldots, e_n\}$.

Worked Example: Construction of an Orthonormal Basis in $\mathbb{R}^3$

Consider the set of linearly independent vectors in $\mathbb{R}^3$:

$$v_1 = (1, 1, 0)$$

$$v_2 = (1, 0, 1)$$

$$v_3 = (0, 1, 1)$$

We apply the Gram-Schmidt technique to construct an orthonormal basis $\{e_1, e_2, e_3\}$.

Construction of e_1

We set the first orthogonal vector equal to v_1:

$$u_1 = v_1 = (1, 1, 0)$$

We compute the norm of u_1:

$$\|u_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{1 + 1 + 0} = \sqrt{2}$$

The first orthonormal vector e_1 is:

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}}(1, 1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

Construction of e_2

We compute the scalar product of u_1 and v_2:

$$\langle u_1, v_2 \rangle = (1)(1) + (1)(0) + (0)(1) = 1 + 0 + 0 = 1$$

We compute the norm squared of u_1:

$$\|u_1\|^2 = (\sqrt{2})^2 = 2$$

We calculate the second orthogonal vector u_2:

$$u_2 = v_2 - \frac{\langle u_1, v_2 \rangle}{\|u_1\|^2} u_1$$

$$u_2 = (1,0,1) - \frac{1}{2}(1,1,0)$$

$$u_2 = \left(1 - \frac{1}{2}, 0 - \frac{1}{2}, 1 - 0\right) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

We compute the norm of u_2:

$$\|u_2\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + 1^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{2}{4} + 1} = \sqrt{\frac{3}{2}}$$

The second orthonormal vector e_2 is:

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{3/2}} \left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \sqrt{\frac{2}{3}} \left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$e_2 = \left(\frac{\sqrt{2}}{2\sqrt{3}}, -\frac{\sqrt{2}}{2\sqrt{3}}, \sqrt{\frac{2}{3}}\right) = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}}\right)$$

Construction of e_3

We compute the scalar product of u_1 and v_3:

$$\langle u_1, v_3 \rangle = (1)(0) + (1)(1) + (0)(1) = 0 + 1 + 0 = 1$$

We compute the scalar product of u_2 and v_3:

$$\langle u_2, v_3 \rangle = \left(\frac{1}{2}\right)(0) + \left(-\frac{1}{2}\right)(1) + (1)(1) = 0 - \frac{1}{2} + 1 = \frac{1}{2}$$

We compute the norm squared of u_2:

$$\|u_2\|^2 = \left(\sqrt{\frac{3}{2}}\right)^2 = \frac{3}{2}$$

We calculate the third orthogonal vector u_3:

$$u_3 = v_3 - \frac{\langle u_1, v_3 \rangle}{\|u_1\|^2} u_1 - \frac{\langle u_2, v_3 \rangle}{\|u_2\|^2} u_2$$

$$u_3 = (0,1,1) - \frac{1}{2}(1,1,0) - \frac{1/2}{3/2} \left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

Simplifying the coefficient of the third term:

$$\frac{1/2}{3/2} = \frac{1}{3}$$

Substituting this scalar value back into u_3:

$$u_3 = (0,1,1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \frac{1}{3}\left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$u_3 = (0,1,1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}\right)$$

Evaluating component-by-component:

$$u_{3,x} = 0 - \frac{1}{2} - \frac{1}{6} = -\frac{3}{6} - \frac{1}{6} = -\frac{4}{6} = -\frac{2}{3}$$

$$u_{3,y} = 1 - \frac{1}{2} - \left(-\frac{1}{6}\right) = \frac{1}{2} + \frac{1}{6} = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

$$u_{3,z} = 1 - 0 - \frac{1}{3} = \frac{2}{3}$$

So the vector u_3 is:

$$u_3 = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

We compute the norm of u_3:

$$\|u_3\| = \sqrt{\left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{12}{9}} = \frac{\sqrt{12}}{3} = \frac{2\sqrt{3}}{3} = \frac{2}{\sqrt{3}}$$

The third orthonormal vector e_3 is:

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{\sqrt{3}}{2}\left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right) = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

Thus, the constructed orthonormal basis for $\mathbb{R}^3$ is:

$$e_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$e_2 = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}}\right)$$

$$e_3 = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

4. Linear Operators and Their Matrix Representation

Definition of a Linear Operator

A linear operator $\hat{A}$ defined on a vector space V over a field F is a mapping that assigns to every vector $|v\rangle \in V$ another vector $\hat{A}|v\rangle \in V$, while preserving the operations of vector addition and scalar multiplication.

The condition of linearity requires that for any vectors $|u\rangle, |v\rangle \in V$ and any scalars $c_1, c_2 \in F$, the operator satisfies:

$$\hat{A}(c_1|u\rangle + c_2|v\rangle) = c_1\hat{A}|u\rangle + c_2\hat{A}|v\rangle$$

This property implies that the operator distributes over vector sums and commutes with scalar multiplication.

Matrix Representation of an Operator

Let $\{|e_1\rangle, |e_2\rangle, \ldots, |e_n\rangle\}$ be an ordered basis for an n-dimensional vector space V. The action of the linear operator $\hat{A}$ on any basis vector $|e_j\rangle$ produces a vector in V, which can be expanded uniquely as a linear combination of the basis vectors:

$$\hat{A}|e_j\rangle = \sum_{i=1}^{n} A_{ij}|e_i\rangle$$

The coefficients A_{ij} are the elements of an $n \times n$ matrix A representing the operator $\hat{A}$ with respect to the chosen basis. Here, the first index i labels the row (corresponding to the basis vector component) and the second index j labels the column (corresponding to the transformed basis vector).

Consider an arbitrary vector $|v\rangle \in V$ expanded in the same basis:

$$|v\rangle = \sum_{j=1}^{n} v_j|e_j\rangle$$

Applying the linear operator $\hat{A}$ to $|v\rangle$ gives:

$$\hat{A}|v\rangle = \hat{A}\left(\sum_{j=1}^{n} v_j|e_j\rangle\right)$$

Using the linearity property of $\hat{A}$, the operator acts directly on the basis vectors inside the sum:

$$\hat{A}|v\rangle = \sum_{j=1}^{n} v_j\hat{A}|e_j\rangle$$

Substituting the expansion of $\hat{A}|e_j\rangle$ into this equation yields:

$$\hat{A}|v\rangle = \sum_{j=1}^{n} v_j \sum_{i=1}^{n} A_{ij}|e_i\rangle$$

Interchanging the order of summation gives:

$$\hat{A}|v\rangle = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} A_{ij} v_j \right) |e_i\rangle$$

If we set $|w\rangle = \hat{A}|v\rangle = \sum_{i=1}^{n} w_i |e_i\rangle$, comparing the coefficients of $|e_i\rangle$ leads to the component relation:

$$w_i = \sum_{j=1}^{n} A_{ij} v_j$$

In matrix notation, this linear system is expressed as matrix multiplication:

$$\mathbf{Y} = A\mathbf{X}$$

where $\mathbf{X}$ and $\mathbf{Y}$ are the column vectors representing $|v\rangle$ and $|w\rangle$, respectively.

Change of Basis and Similarity Transformation

Suppose we change from an old basis $\{|e_i\rangle\}$ to a new basis $\{|e'_i\rangle\}$. Each vector of the new basis can be expanded in terms of the old basis:

$$|e'_j\rangle = \sum_{i=1}^{n} S_{ij} |e_i\rangle$$

Here, S is the $n \times n$ transformation matrix whose j-th column contains the components of $|e'_j\rangle$ in the old basis. The inverse transformation expresses the old basis vectors in terms of the new basis:

$$|e_j\rangle = \sum_{i=1}^{n} (S^{-1})_{ij} |e'_i\rangle$$

Let A' be the matrix representation of the operator $\hat{A}$ in the new basis, defined by:

$$\hat{A}|e'_j\rangle = \sum_{k=1}^{n} A'_{kj} |e'_k\rangle$$

Substituting the transformation relation for $|e'_j\rangle$ into the left-hand side gives:

$$\hat{A}|e'_j\rangle = \hat{A} \left(\sum_{m=1}^{n} S_{mj} |e_m\rangle \right)$$

By linearity, the operator acts on the basis vectors inside the summation:

$$\hat{A}|e'_j\rangle = \sum_{m=1}^{n} S_{mj}\hat{A}|e_m\rangle$$

Substituting the action of $\hat{A}$ on the old basis vectors yields:

$$\hat{A}|e'_j\rangle = \sum_{m=1}^{n} S_{mj}\sum_{i=1}^{n} A_{im}|e_i\rangle$$

Now expressing the old basis vectors $|e_i\rangle$ in terms of the new basis vectors using S^{-1} gives:

$$\hat{A}|e'_j\rangle = \sum_{m=1}^{n} S_{mj}\sum_{i=1}^{n} A_{im}\sum_{k=1}^{n}(S^{-1})_{ki}|e'_k\rangle$$

Reordering the summations produces:

$$\hat{A}|e'_j\rangle = \sum_{k=1}^{n}\left(\sum_{i=1}^{n}\sum_{m=1}^{n}(S^{-1})_{ki}A_{im}S_{mj}\right)|e'_k\rangle$$

Comparing this with the definition $\hat{A}|e'_j\rangle = \sum_{k=1}^{n} A'_{kj}|e'_k\rangle$, the components of the matrix A' are:

$$A'_{kj} = \sum_{i=1}^{n}\sum_{m=1}^{n}(S^{-1})_{ki}A_{im}S_{mj}$$

In matrix notation, this relationship is written as:

$$A' = S^{-1}AS$$

This transformation of a matrix representation under a change of basis is called a similarity transformation, and matrices A and A' are said to be similar.

Singular and Non-Singular Matrices and Matrix Inverse

A square matrix A of dimension $n \times n$ is classified based on its determinant: - A matrix A is called **non-singular** if its determinant is non-zero, i.e., $\det(A) \neq 0$. - A matrix A is called **singular** if its determinant is zero, i.e., $\det(A) = 0$.

For a non-singular matrix A, there exists a unique matrix called the inverse, denoted by A^{-1}, such that:

$$AA^{-1} = A^{-1}A = I$$

where I is the $n \times n$ identity matrix.

The inverse matrix can be explicitly calculated using the adjugate matrix:

$$A^{-1} = \frac{1}{\det(A)}\mathrm{adj}(A)$$

where $\mathrm{adj}(A)$ is the transpose of the cofactor matrix of A. If $\det(A) = 0$, the inverse does not exist.

5. Eigenvalues and Eigenvectors

Characteristic Polynomial and Characteristic Equation

Let A be an $n \times n$ matrix representing a linear operator. A non-zero column vector X is called an eigenvector of A if operating A on X results in a scalar multiple of X:

$$AX = \lambda X$$

where λ is a scalar called the eigenvalue corresponding to the eigenvector X.

Subtracting λX from both sides gives:

$$AX - \lambda X = \mathbf{0}$$

Inserting the identity matrix I allows factoring out the vector X:

$$(A - \lambda I)X = \mathbf{0}$$

This is a system of homogeneous linear equations. A non-trivial solution $X \neq \mathbf{0}$ exists if and only if the matrix coefficient $(A - \lambda I)$ is singular. Therefore, the determinant must vanish:

$$\det(A - \lambda I) = 0$$

Expanding this determinant yields an n-th degree polynomial in λ:

$$P(\lambda) = \det(A - \lambda I)$$

This polynomial $P(\lambda)$ is called the **characteristic polynomial** of A. The equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A. The roots of the characteristic equation are the eigenvalues of A.

Worked Example: Eigenvalues and Eigenvectors of a 3x3 Matrix

Consider the 3×3 matrix:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & 1 & 3 \end{pmatrix}$$

To find the eigenvalues, we set up the matrix $(A - \lambda I)$:

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & 1 & 3-\lambda \end{pmatrix}$$

The characteristic equation is given by evaluating the determinant:

$$\det(A - \lambda I) = 0$$

Expanding the determinant along the first row yields:

$$\det(A-\lambda I) = (1-\lambda)\det\begin{pmatrix}2-\lambda & 0\\ 1 & 3-\lambda\end{pmatrix} - 2\det\begin{pmatrix}0 & 0\\ -2 & 3-\lambda\end{pmatrix} + 0$$

Evaluating the 2×2 determinants gives:

$$\det(A-\lambda I) = (1-\lambda)[(2-\lambda)(3-\lambda)-0] - 0$$

Factoring the resulting expression yields the characteristic polynomial equation:

$$(1-\lambda)(2-\lambda)(3-\lambda) = 0$$

The roots of this equation are the three distinct eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3$$

Now we find the eigenvector corresponding to each eigenvalue.

Eigenvector for $\lambda_1 = 1$:

The eigenvector equation is:

$$(A - 1I)X_1 = \mathbf{0}$$

Substituting $\lambda = 1$ into $(A-\lambda I)$ gives:

$$\begin{pmatrix}0 & 2 & 0\\ 0 & 1 & 0\\ -2 & 1 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

Writing out the linear system of equations:

$$2x_2 = 0$$

$$x_2 = 0$$

$$-2x_1 + x_2 + 2x_3 = 0$$

From the first two equations, we have $x_2 = 0$. Substituting $x_2 = 0$ into the third equation gives:

$$-2x_1 + 2x_3 = 0$$

Rearranging terms yields:

$$x_1 = x_3$$

Choosing $x_3 = 1$ gives $x_1 = 1$. Thus, the eigenvector corresponding to $\lambda_1 = 1$ is:

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Eigenvector for $\lambda_2 = 2$:

The eigenvector equation is:

$$(A - 2I)X_2 = \mathbf{0}$$

Substituting $\lambda = 2$ into $(A - \lambda I)$ gives:

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Writing out the system of equations:

$$-x_1 + 2x_2 = 0$$

$$-2x_1 + x_2 + x_3 = 0$$

From the first equation, we express x_1 in terms of x_2:

$$x_1 = 2x_2$$

Substituting $x_1 = 2x_2$ into the second equation yields:

$$-2(2x_2) + x_2 + x_3 = 0$$

Combining like terms gives:

$$-3x_2 + x_3 = 0$$

Rearranging terms yields:

$$x_3 = 3x_2$$

Choosing $x_2 = 1$, we find $x_1 = 2$ and $x_3 = 3$. Thus, the eigenvector corresponding to $\lambda_2 = 2$ is:

$$X_2 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

Eigenvector for $\lambda_3 = 3$:

The eigenvector equation is:

$$(A - 3I)X_3 = \mathbf{0}$$

Substituting $\lambda = 3$ into $(A - \lambda I)$ gives:

$$\begin{pmatrix} -2 & 2 & 0 \\ 0 & -1 & 0 \\ -2 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Writing out the system of equations:

$$-2x_1 + 2x_2 = 0$$

$$-x_2 = 0$$

$$-2x_1 + x_2 = 0$$

From the second equation, we get $x_2 = 0$. Substituting $x_2 = 0$ into the first equation yields:

$$-2x_1 = 0 \implies x_1 = 0$$

The component x_3 is arbitrary. Choosing $x_3 = 1$, the eigenvector corresponding to $\lambda_3 = 3$ is:

$$X_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

6. Cayley-Hamilton Theorem

Statement of Cayley-Hamilton Theorem

The Cayley-Hamilton Theorem states that every square matrix satisfies its own characteristic equation. If the characteristic polynomial of an $n \times n$ matrix A is given by:

$$P(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 = 0$$

Then replacing the scalar variable λ with the matrix A (and replacing c_0 with $c_0 I$) yields the matrix equation:

$$P(A) = (-1)^n A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = \mathbf{0}$$

where $\mathbf{0}$ represents the $n \times n$ zero matrix.

Proof of Cayley-Hamilton Theorem for a 2x2 Matrix

Let A be an arbitrary 2×2 matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The characteristic equation of A is given by:

$$\det(A - \lambda I) = 0$$

Expanding this determinant explicitly:

$$\det\begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = 0$$

Calculating the 2×2 determinant yields:

$$(a-\lambda)(d-\lambda) - bc = 0$$

Multiplying out the product gives:

$$ad - a\lambda - d\lambda + \lambda^2 - bc = 0$$

Rearranging terms in powers of λ yields:

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0$$

We identify the trace $\mathrm{Tr}(A) = a + d$ and the determinant $\det(A) = ad - bc$. The characteristic polynomial is:

$$P(\lambda) = \lambda^2 - \mathrm{Tr}(A)\lambda + \det(A)$$

To prove the Cayley-Hamilton theorem, we must verify that substituting A for λ yields the zero matrix:

$$P(A) = A^2 - (a+d)A + (ad-bc)I = \mathbf{0}$$

First, we calculate A^2:

$$A^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Performing matrix multiplication gives:

$$A^2 = \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}$$

Next, we evaluate the term $(a+d)A$:

$$(a+d)A = (a+d)\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a(a+d) & b(a+d) \\ c(a+d) & d(a+d) \end{pmatrix}$$

Expanding each entry gives:

$$(a+d)A = \begin{pmatrix} a^2+ad & ab+bd \\ ac+cd & ad+d^2 \end{pmatrix}$$

We also write the scalar multiple of the identity matrix $(ad-bc)I$:

$$(ad-bc)I = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$

Now, substituting A^2, $(a+d)A$, and $(ad-bc)I$ into $P(A)$ gives:

$$P(A) = \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix} - \begin{pmatrix} a^2+ad & ab+bd \\ ac+cd & ad+d^2 \end{pmatrix} + \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$

Combining all three matrices entry by entry:

$$P(A) = \begin{pmatrix} (a^2+bc)-(a^2+ad)+(ad-bc) & (ab+bd)-(ab+bd)+0 \\ (ac+cd)-(ac+cd)+0 & (bc+d^2)-(ad+d^2)+(ad-bc) \end{pmatrix}$$

Simplifying the algebraic expressions inside each entry yields:

$$P(A) = \begin{pmatrix} a^2+bc-a^2-ad+ad-bc & ab+bd-ab-bd \\ ac+cd-ac-cd & bc+d^2-ad-d^2+ad-bc \end{pmatrix}$$

Cancelling equal and opposite terms in all entries results in:

$$P(A) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{0}$$

This completes the proof of the Cayley-Hamilton theorem for a 2×2 matrix.

Statement and Application of Cayley-Hamilton Theorem for a 3x3 Matrix

For a general 3×3 matrix A, the characteristic equation is:

$$\det(A - \lambda I) = 0$$

Expanding this determinant yields a cubic characteristic polynomial:

$$\lambda^3 - p\lambda^2 + q\lambda - r = 0$$

where the coefficients are given by $p = \mathrm{Tr}(A)$, $q = \frac{1}{2}\left[(\mathrm{Tr}(A))^2 - \mathrm{Tr}(A^2)\right]$, and $r = \det(A)$.

By the Cayley-Hamilton theorem, substituting matrix A into this equation gives:

$$A^3 - pA^2 + qA - rI = \mathbf{0}$$

This theorem allows higher powers of a matrix A (such as $A^3, A^4, \ldots$) to be expressed as linear combinations of lower powers (I, A, A^2).

Finding Matrix Inverse Using Cayley-Hamilton Theorem

The Cayley-Hamilton theorem provides a method to compute the inverse of a non-singular matrix. Assume A is an $n \times n$ non-singular matrix, so that $\det(A) \neq 0$.

For a 3×3 matrix, the Cayley-Hamilton relation is:

$$A^3 - pA^2 + qA - rI = \mathbf{0}$$

where $r = \det(A) \neq 0$.

Multiplying the entire equation from the left by A^{-1} yields:

$$A^{-1}\left(A^3 - pA^2 + qA - rI\right) = \mathbf{0}$$

Distributing A^{-1} across the terms inside the parentheses gives:

$$A^{-1}A^3 - pA^{-1}A^2 + qA^{-1}A - rA^{-1}I = \mathbf{0}$$

Using the identity $A^{-1}A = I$, this simplifies to:

$$A^2 - pA + qI - rA^{-1} = \mathbf{0}$$

Isolating the term containing A^{-1} on one side gives:

$$rA^{-1} = A^2 - pA + qI$$

Dividing by $r = \det(A)$ yields the explicit expression for the inverse matrix:

$$A^{-1} = \frac{1}{\det(A)}\left(A^2 - pA + qI\right)$$

7. Orthogonal Matrices

Transpose and Conjugate Transpose

Let A be an $m \times n$ matrix with real or complex entries A_{ij}. The transpose of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A. The elements of the transpose matrix are defined by

$$(A^T)_{ij} = A_{ji}$$

The conjugate transpose (also called the Hermitian conjugate or adjoint) of a matrix A, denoted by $A^\dagger$, is obtained by taking the complex conjugate of each entry of the transpose matrix. Its entries are defined by

$$(A^\dagger)_{ij} = (A_{ji})^*$$

For a real matrix, all entries are real, so the conjugate transpose reduces directly to the matrix transpose, $A^\dagger = A^T$.

Symmetric and Antisymmetric Matrices

A real square matrix A of order $n \times n$ is said to be symmetric if it is equal to its transpose:

$$A^T = A$$

In terms of components, a symmetric matrix satisfies $A_{ij} = A_{ji}$ for all indices $i, j = 1, 2, \ldots, n$.

A real square matrix A of order $n \times n$ is said to be antisymmetric (or skew-symmetric) if it is equal to the negative of its transpose:

$$A^T = -A$$

In terms of components, an antisymmetric matrix satisfies $A_{ij} = -A_{ji}$ for all indices i, j. Setting $i = j$, the diagonal elements satisfy $A_{ii} = -A_{ii}$, which implies that all diagonal elements of an antisymmetric matrix are zero:

$$A_{ii} = 0$$

Unique Decomposition of a Square Matrix

Every real square matrix A can be uniquely expressed as the sum of a symmetric matrix S and an antisymmetric matrix K.

To prove existence, let us construct two matrices S and K from A as follows:

$$S = \frac{1}{2}(A + A^T)$$

$$K = \frac{1}{2}(A - A^T)$$

Taking the transpose of the matrix S, we apply the distributive property of the transpose operation:

$$S^T = \left[\frac{1}{2}(A + A^T)\right]^T$$

Using the identity $(B + C)^T = B^T + C^T$ and $(A^T)^T = A$, this evaluates to

$$S^T = \frac{1}{2}(A^T + A)$$

Since matrix addition is commutative, we obtain

$$S^T = \frac{1}{2}(A + A^T) = S$$

Thus, S is a symmetric matrix.

Next, taking the transpose of the matrix K, we write

$$K^T = \left[\frac{1}{2}(A - A^T)\right]^T$$

Applying the linearity of the transpose operation yields

$$K^T = \frac{1}{2}(A^T - A)$$

Factoring out a negative sign from the right-hand side gives

$$K^T = -\frac{1}{2}(A - A^T)$$

Substituting the definition of K, we find

$$K^T = -K$$

Thus, K is an antisymmetric matrix.

Adding S and K together gives

$$S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Combining terms over a common denominator yields

$$S + K = \frac{1}{2}(A + A^T + A - A^T)$$

Simplifying the expression inside the parentheses results in

$$S + K = \frac{1}{2}(2A) = A$$

This proves that any square matrix A can be expressed as the sum of a symmetric matrix and an antisymmetric matrix.

To prove uniqueness, suppose there exists another decomposition such that

$$A = S' + K'$$

where S' is symmetric ($S'^T = S'$) and K' is antisymmetric ($K'^T = -K'$). Taking the transpose of both sides of this decomposition gives

$$A^T = (S' + K')^T$$

Applying the sum rule for transposes yields

$$A^T = S'^T + K'^T$$

Substituting the symmetry condition $S'^T = S'$ and antisymmetry condition $K'^T = -K'$ gives

$$A^T = S' - K'$$

Adding the expression for A to the expression for A^T yields

$$A + A^T = (S' + K') + (S' - K')$$

Simplifying the right-hand side gives

$$A + A^T = 2S'$$

Dividing both sides by 2 yields

$$S' = \frac{1}{2}(A + A^T) = S$$

Subtracting A^T from A yields

$$A - A^T = (S' + K') - (S' - K')$$

Simplifying the right-hand side gives

$$A - A^T = 2K'$$

Dividing both sides by 2 yields

$$K' = \frac{1}{2}(A - A^T) = K$$

Since $S' = S$ and $K' = K$, the decomposition is unique.

Definition and Properties of Orthogonal Matrices

A real square matrix O of order $n \times n$ is defined as orthogonal if its product with its transpose yields the identity matrix I:

$$O^T O = O O^T = I$$

This definition implies that an orthogonal matrix is invertible, and its inverse is equal to its transpose:

$$O^{-1} = O^T$$

Preservation of Scalar Products

An orthogonal matrix preserves the standard inner product of vectors under transformation. Let $\mathbf{x}$ and $\mathbf{y}$ be real column vectors in $\mathbb{R}^n$, with scalar product defined by $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$.

Let transformed vectors be defined by $\mathbf{x}' = O\mathbf{x}$ and $\mathbf{y}' = O\mathbf{y}$. The scalar product of the transformed vectors is

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (\mathbf{x}')^T \mathbf{y}'$$

Substituting the linear transformations $\mathbf{x}' = O\mathbf{x}$ and $\mathbf{y}' = O\mathbf{y}$ gives

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (O\mathbf{x})^T (O\mathbf{y})$$

Using the matrix transpose identity $(AB)^T = B^T A^T$, the first term expands to

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^T O^T O \mathbf{y}$$

Using the orthogonality condition $O^T O = I$, this becomes

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^T I \mathbf{y}$$

Multiplying by the identity matrix leaves the vector unchanged:

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^T \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$$

Thus, orthogonal transformations preserve scalar products, which guarantees that vector lengths and angles between vectors are preserved.

Column Vectors Form an Orthonormal Basis

Let the $n \times n$ orthogonal matrix O be written in terms of its column vectors as

$$O = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n \end{bmatrix}$$

The transpose matrix O^T has these vectors as its rows:

$$O^T = \begin{bmatrix} \mathbf{c}_1^T \\ \mathbf{c}_2^T \\ \vdots \\ \mathbf{c}_n^T \end{bmatrix}$$

The product $O^T O$ is then given by

$$O^T O = \begin{bmatrix} \mathbf{c}_1^T \mathbf{c}_1 & \mathbf{c}_1^T \mathbf{c}_2 & \dots & \mathbf{c}_1^T \mathbf{c}_n \\ \mathbf{c}_2^T \mathbf{c}_1 & \mathbf{c}_2^T \mathbf{c}_2 & \dots & \mathbf{c}_2^T \mathbf{c}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{c}_n^T \mathbf{c}_1 & \mathbf{c}_n^T \mathbf{c}_2 & \dots & \mathbf{c}_n^T \mathbf{c}_n \end{bmatrix}$$

Since $O^T O = I$, the (i, j)-entry of this matrix product must equal the Kronecker delta δ_{ij}:

$$\mathbf{c}_i^T \mathbf{c}_j = \delta_{ij}$$

When $i = j$, we have $\mathbf{c}_i^T \mathbf{c}_i = 1$, meaning each column vector has unit norm. When $i \neq j$, we have $\mathbf{c}_i^T \mathbf{c}_j = 0$, meaning distinct column vectors are orthogonal. Therefore, the column vectors of an orthogonal matrix form an orthonormal basis for $\mathbb{R}^n$.

8. Unitary Matrices

Definition of Unitary Matrices

A complex square matrix U of order $n \times n$ is said to be unitary if its product with its conjugate transpose $U^\dagger$ equals the identity matrix:

$$U^\dagger U = U U^\dagger = I$$

This condition implies that a unitary matrix is invertible, with its inverse matrix given by its conjugate transpose:

$$U^{-1} = U^\dagger$$

Unitary matrices represent the complex generalization of real orthogonal matrices.

Properties of Unitary Matrices

Preservation of Complex Inner Product

Let $\mathbf{x}$ and $\mathbf{y}$ be complex vectors in $\mathbb{C}^n$, with inner product defined by $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\dagger \mathbf{y}$. Under a transformation by a unitary matrix U, the transformed vectors are $\mathbf{x}' = U\mathbf{x}$ and $\mathbf{y}' = U\mathbf{y}$.

The inner product of the transformed vectors is written as

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (\mathbf{x}')^\dagger \mathbf{y}'$$

Substituting the transformed expressions gives

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (U\mathbf{x})^\dagger (U\mathbf{y})$$

Applying the matrix property $(AB)^\dagger = B^\dagger A^\dagger$, this expands to

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^\dagger U^\dagger U \mathbf{y}$$

Applying the unitarity condition $U^\dagger U = I$ simplifies the expression to

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^\dagger I \mathbf{y}$$

Multiplying by the identity matrix gives

$$\langle \mathbf{x}', \mathbf{y}' \rangle = \mathbf{x}^\dagger \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$$

Hence, unitary operators preserve the inner product and norm of vectors in complex vector spaces.

Determinant of a Unitary Matrix

Taking the determinant of both sides of the relation $U^\dagger U = I$ gives

$$\det(U^\dagger U) = \det(I)$$

Using the property $\det(AB) = \det(A)\det(B)$ and $\det(I) = 1$, we write

$$\det(U^\dagger)\det(U) = 1$$

Since the determinant of the conjugate transpose is the complex conjugate of the determinant, $\det(U^\dagger) = (\det(U))^*$, this equation becomes

$$(\det(U))^* \det(U) = 1$$

Using the property of complex numbers $z^* z = |z|^2$, we obtain

$$|\det(U)|^2 = 1$$

Taking the positive square root gives

$$|\det(U)| = 1$$

Thus, the determinant of any unitary matrix has absolute value equal to one.

Eigenvalues of a Unitary Matrix Have Modulus One

Let λ be an eigenvalue of a unitary matrix U, with corresponding non-zero eigenvector $\mathbf{x}$ satisfying the eigenvalue equation

$$U\mathbf{x} = \lambda\mathbf{x}$$

Taking the conjugate transpose of both sides of this eigenvalue equation gives

$$(U\mathbf{x})^\dagger = (\lambda\mathbf{x})^\dagger$$

Applying the conjugate transpose property $(AB)^\dagger = B^\dagger A^\dagger$ to the left-hand side and $(\alpha\mathbf{x})^\dagger = \alpha^*\mathbf{x}^\dagger$ to the right-hand side yields

$$\mathbf{x}^\dagger U^\dagger = \lambda^*\mathbf{x}^\dagger$$

Multiplying the left-hand side of $\mathbf{x}^\dagger U^\dagger = \lambda^*\mathbf{x}^\dagger$ by the left-hand side of $U\mathbf{x} = \lambda\mathbf{x}$ gives

$$(\mathbf{x}^\dagger U^\dagger)(U\mathbf{x}) = (\lambda^*\mathbf{x}^\dagger)(\lambda\mathbf{x})$$

Regrouping terms using matrix associativity yields

$$\mathbf{x}^\dagger(U^\dagger U)\mathbf{x} = \lambda^*\lambda(\mathbf{x}^\dagger\mathbf{x})$$

Substituting the unitarity condition $U^\dagger U = I$ into the left-hand side yields

$$\mathbf{x}^\dagger I\mathbf{x} = |\lambda|^2(\mathbf{x}^\dagger\mathbf{x})$$

Simplifying the left-hand side gives

$$\mathbf{x}^\dagger\mathbf{x} = |\lambda|^2(\mathbf{x}^\dagger\mathbf{x})$$

Rearranging all terms to one side yields

$$(1 - |\lambda|^2)(\mathbf{x}^\dagger\mathbf{x}) = 0$$

Since $\mathbf{x}$ is a non-zero eigenvector, its norm square $\mathbf{x}^\dagger\mathbf{x}$ is strictly positive ($\mathbf{x}^\dagger\mathbf{x} > 0$). Dividing by $\mathbf{x}^\dagger\mathbf{x}$ gives

$$1 - |\lambda|^2 = 0$$

Solving for $|\lambda|^2$ gives

$$|\lambda|^2 = 1$$

Taking the positive square root gives

$$|\lambda| = 1$$

This proves that all eigenvalues of a unitary matrix have unit modulus.

9. Hermitian Matrices

Definition of Hermitian Matrix

A complex square matrix H of order $n \times n$ is defined to be Hermitian (or self-adjoint) if it is equal to its conjugate transpose:

$$H^\dagger = H$$

In terms of individual matrix elements, a Hermitian matrix satisfies

$$H_{ij} = (H_{ji})^*$$

Setting $i = j$, the diagonal elements satisfy $H_{ii} = (H_{ii})^*$, which implies that all diagonal elements of a Hermitian matrix must be purely real numbers.

A matrix K is skew-Hermitian if $K^\dagger = -K$, which implies its diagonal elements are purely imaginary or zero.

Real Eigenvalues of a Hermitian Matrix

Let H be a Hermitian matrix, and let λ be an eigenvalue of H with a non-zero eigenvector $\mathbf{x} \in \mathbb{C}^n$. The eigenvalue equation is

$$H\mathbf{x} = \lambda\mathbf{x}$$

Premultiplying both sides of the eigenvalue equation by the row vector $\mathbf{x}^\dagger$ gives

$$\mathbf{x}^\dagger H\mathbf{x} = \mathbf{x}^\dagger(\lambda\mathbf{x})$$

Factoring out the scalar λ from the right-hand side yields

$$\mathbf{x}^\dagger H\mathbf{x} = \lambda(\mathbf{x}^\dagger\mathbf{x})$$

Now, take the conjugate transpose of both sides of the original eigenvalue equation $H\mathbf{x} = \lambda\mathbf{x}$:

$$(H\mathbf{x})^\dagger = (\lambda\mathbf{x})^\dagger$$

Applying the conjugate transpose operation rule $(AB)^\dagger = B^\dagger A^\dagger$ gives

$$\mathbf{x}^\dagger H^\dagger = \lambda^* \mathbf{x}^\dagger$$

Since H is a Hermitian matrix, we substitute $H^\dagger = H$ into this equation:

$$\mathbf{x}^\dagger H = \lambda^* \mathbf{x}^\dagger$$

Post-multiplying both sides of this equation by the column vector $\mathbf{x}$ yields

$$(\mathbf{x}^\dagger H)\mathbf{x} = (\lambda^* \mathbf{x}^\dagger)\mathbf{x}$$

Factoring out the scalar λ^* gives

$$\mathbf{x}^\dagger H \mathbf{x} = \lambda^* (\mathbf{x}^\dagger \mathbf{x})$$

We now have two expressions for $\mathbf{x}^\dagger H \mathbf{x}$. Equating these two expressions gives

$$\lambda (\mathbf{x}^\dagger \mathbf{x}) = \lambda^* (\mathbf{x}^\dagger \mathbf{x})$$

Subtracting the right-hand side from the left-hand side yields

$$(\lambda - \lambda^*)(\mathbf{x}^\dagger \mathbf{x}) = 0$$

Because $\mathbf{x}$ is a non-zero eigenvector, its inner product with itself is positive:

$$\mathbf{x}^\dagger \mathbf{x} > 0$$

Dividing the equation by the non-zero quantity $\mathbf{x}^\dagger \mathbf{x}$ gives

$$\lambda - \lambda^* = 0$$

This implies

$$\lambda = \lambda^*$$

A complex number equal to its complex conjugate must be real. Therefore, all eigenvalues of a Hermitian matrix are real.

Orthogonality of Eigenvectors for Distinct Eigenvalues

Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be eigenvectors of a Hermitian matrix H corresponding to distinct eigenvalues λ_1 and λ_2 respectively, such that $\lambda_1 \neq \lambda_2$.

The eigenvalue relations are given by

$$H\mathbf{x}_1 = \lambda_1 \mathbf{x}_1$$

$$H\mathbf{x}_2 = \lambda_2 \mathbf{x}_2$$

Since H is Hermitian, its eigenvalues λ_1 and λ_2 are real numbers ($\lambda_1^* = \lambda_1$ and $\lambda_2^* = \lambda_2$).

Premultiplying the equation $H\mathbf{x}_2 = \lambda_2 \mathbf{x}_2$ by $\mathbf{x}_1^\dagger$ gives

$$\mathbf{x}_1^\dagger H \mathbf{x}_2 = \mathbf{x}_1^\dagger (\lambda_2 \mathbf{x}_2)$$

Factoring out the scalar λ_2 yields

$$\mathbf{x}_1^\dagger H \mathbf{x}_2 = \lambda_2 (\mathbf{x}_1^\dagger \mathbf{x}_2)$$

Next, taking the conjugate transpose of $H\mathbf{x}_1 = \lambda_1 \mathbf{x}_1$ yields

$$(H\mathbf{x}_1)^\dagger = (\lambda_1 \mathbf{x}_1)^\dagger$$

Applying conjugate transpose rules gives

$$\mathbf{x}_1^\dagger H^\dagger = \lambda_1^* \mathbf{x}_1^\dagger$$

Using the Hermitian property $H^\dagger = H$ and reality of eigenvalues $\lambda_1^* = \lambda_1$, this becomes

$$\mathbf{x}_1^\dagger H = \lambda_1 \mathbf{x}_1^\dagger$$

Post-multiplying this equation by $\mathbf{x}_2$ gives

$$(\mathbf{x}_1^\dagger H)\mathbf{x}_2 = (\lambda_1 \mathbf{x}_1^\dagger)\mathbf{x}_2$$

Factoring out the scalar λ_1 yields

$$\mathbf{x}_1^\dagger H \mathbf{x}_2 = \lambda_1 (\mathbf{x}_1^\dagger \mathbf{x}_2)$$

Equating the two derived expressions for $\mathbf{x}_1^\dagger H \mathbf{x}_2$ gives

$$\lambda_1 (\mathbf{x}_1^\dagger \mathbf{x}_2) = \lambda_2 (\mathbf{x}_1^\dagger \mathbf{x}_2)$$

Rearranging terms to one side yields

$$(\lambda_1 - \lambda_2)(\mathbf{x}_1^\dagger \mathbf{x}_2) = 0$$

By hypothesis, the eigenvalues are distinct, so $\lambda_1 \neq \lambda_2$, which implies

$$\lambda_1 - \lambda_2 \neq 0$$

Dividing by the non-zero scalar $(\lambda_1 - \lambda_2)$ gives

$$\mathbf{x}_1^\dagger \mathbf{x}_2 = 0$$

Thus, the eigenvectors $\mathbf{x}_1$ and $\mathbf{x}_2$ corresponding to distinct eigenvalues of a Hermitian matrix are orthogonal.

10. Infinite Dimensional Spaces

The concept of a vector space can be extended beyond finite dimensions to spaces with an infinite number of linearly independent vectors. In an infinite-dimensional vector space, a vector is specified by an infinite sequence of components rather than a finite set of n coordinates.

A primary example of an infinite-dimensional vector space is the space of square-summable sequences, denoted by l^2. A vector x in l^2 is represented as an infinite ordered sequence of numbers:

$$x = (x_1, x_2, x_3, \dots)$$

where each component x_i is a real or complex number. For this sequence to belong to the space l^2, it must satisfy the square-summability condition, which requires the infinite series of the squared absolute values of its components to converge to a finite value:

$$\sum_{i=1}^{\infty} |x_i|^2 < \infty$$

The space l^2 satisfies all vector space axioms under component-wise addition and scalar multiplication. For two vectors $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ in l^2, vector addition is defined as:

$$x + y = (x_1 + y_1, x_2 + y_2, x_3 + y_3, \dots)$$

For any complex or real scalar α, scalar multiplication is defined as:

$$\alpha x = (\alpha x_1, \alpha x_2, \alpha x_3, \dots)$$

An infinite-dimensional space requires an infinite basis to span every vector in the space. In l^2, the standard canonical basis consists of an infinite set of vectors $\{e_1, e_2, e_3, \dots\}$, where the k-th basis vector e_k has 1 in the k-th entry and zeros in all other entries:

$$e_1 = (1, 0, 0, 0, \dots)$$

$$e_2 = (0, 1, 0, 0, \dots)$$

$$e_k = (0, 0, \dots, 1, 0, \dots)$$

This infinite collection of basis vectors is linearly independent, as no element in the set can be written as a finite linear combination of the other elements.

To represent vectors as infinite linear combinations of basis vectors, a clear definition of convergence in a vector space is required. Let $\{x^{(n)}\}$ be a sequence of vectors $x^{(1)}, x^{(2)}, x^{(3)}, \dots$ in a normed infinite-dimensional space. The sequence of vectors $\{x^{(n)}\}$ converges to a limit vector x if the norm of their difference approaches zero as n approaches infinity:

$$\lim_{n \to \infty} \|x^{(n)} - x\| = 0$$

Using vector convergence, an infinite series of vectors $\sum_{k=1}^{\infty} v_k$ is defined through its sequence of partial sums. The n-th partial sum vector s_n is the finite sum of the first n vector terms:

$$s_n = \sum_{k=1}^{n} v_k$$

The infinite series $\sum_{k=1}^{\infty} v_k$ converges to a target vector v if the sequence of partial sums $\{s_n\}$ converges to v in norm:

$$\lim_{n \to \infty} \|s_n - v\| = 0$$

This requires that the distance between the partial sum s_n and the vector v can be made arbitrarily small by taking a sufficiently large number of terms n.

11. Hilbert Space

A Hilbert space, denoted by $\mathcal{H}$, is a vector space equipped with an inner product that is complete with respect to the norm induced by that inner product.

An inner product on a complex vector space $\mathcal{H}$ assigns a scalar $\langle u, v \rangle$ to every pair of vectors $u, v \in \mathcal{H}$. It satisfies conjugate symmetry:

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

It satisfies linearity in its first argument for any scalar α:

$$\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$$

It is conjugate-linear in its second argument:

$$\langle u, \alpha v \rangle = \bar{\alpha} \langle u, v \rangle$$

It satisfies distributivity over vector addition:

$$\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$$

It satisfies positive-definiteness, where the inner product of a vector with itself is non-negative and vanishes only for the zero vector:

$$\langle u, u \rangle \geq 0, \quad \text{with } \langle u, u \rangle = 0 \iff u = 0$$

The inner product induces a natural norm on the Hilbert space $\mathcal{H}$, defining the length or magnitude of any vector $u \in \mathcal{H}$:

$$\|u\| = \sqrt{\langle u, u \rangle}$$

Completeness is the defining characteristic that separates a general inner product space from a Hilbert space. A sequence of vectors $\{x_n\}$ in $\mathcal{H}$ is called a Cauchy sequence if the norm of the difference between its terms approaches zero as the indices go to infinity:

$$\lim_{n,m \to \infty} \|x_n - x_m\| = 0$$

A space $\mathcal{H}$ is complete if every Cauchy sequence $\{x_n\}$ constructed from elements in $\mathcal{H}$ converges to a limit vector x that is also a member of $\mathcal{H}$:

$$\lim_{n\to\infty} \|x_n - x\| = 0, \quad \text{where } x \in \mathcal{H}$$

In a Hilbert space, a countable set of vectors $\{\phi_1, \phi_2, \dots\}$ forms an orthonormal set if all vectors in the set are mutually orthogonal and normalized to unit length:

$$\langle \phi_i, \phi_j \rangle = \delta_{ij}$$

where δ_{ij} is the Kronecker delta symbol. The orthonormal set forms an orthonormal basis for $\mathcal{H}$ if the only vector orthogonal to every vector in the set is the zero vector.

Any arbitrary vector $x \in \mathcal{H}$ can be expanded as an infinite linear combination of the orthonormal basis vectors:

$$x = \sum_{i=1}^{\infty} c_i \phi_i$$

To determine the explicit expression for the expansion coefficients c_i, we take the inner product of the vector x with a specific basis vector ϕ_j:

$$\langle x, \phi_j \rangle = \left\langle \sum_{i=1}^{\infty} c_i \phi_i, \phi_j \right\rangle$$

Exploiting the linearity of the inner product in its first argument, the summation sign and the scalar coefficients c_i can be moved outside the inner product bracket:

$$\langle x, \phi_j \rangle = \sum_{i=1}^{\infty} c_i \langle \phi_i, \phi_j \rangle$$

Substituting the orthonormality condition $\langle \phi_i, \phi_j \rangle = \delta_{ij}$ into the right-hand side yields:

$$\langle x, \phi_j \rangle = \sum_{i=1}^{\infty} c_i \delta_{ij}$$

Evaluating the sum over the Kronecker delta collapses all terms except the single term where $i = j$:

$$\langle x, \phi_j \rangle = c_j$$

Thus, each coefficient c_i in the series expansion is equal to the inner product of the vector x with the corresponding basis vector ϕ_i:

$$c_i = \langle x, \phi_i \rangle$$

Substituting this coefficient back into the expansion gives the exact representation of x:

$$x = \sum_{i=1}^{\infty} \langle x, \phi_i \rangle \phi_i$$

Completeness of the Hilbert space ensures that infinite series representations of vectors are mathematically valid and fully well-behaved. When a vector is expanded into an infinite series, the partial sums $s_n = \sum_{i=1}^{n} c_i \phi_i$ form a sequence of vectors. Completeness guarantees that if this sequence of partial sums forms a Cauchy sequence, its limit vector x is guaranteed to exist inside the space $\mathcal{H}$ rather than outside it.

Furthermore, completeness ensures that the norm of any vector x can be computed directly from its expansion coefficients, a relation known as Parseval's identity. To derive Parseval's identity, we express the squared norm of x using its inner product definition:

$$\|x\|^2 = \langle x, x \rangle$$

Substituting the infinite series expansion for x into both arguments of the inner product gives:

$$\|x\|^2 = \left\langle \sum_{i=1}^{\infty} c_i \phi_i, \sum_{j=1}^{\infty} c_j \phi_j \right\rangle$$

Using linearity in the first argument and conjugate-linearity in the second argument allows the infinite sums to be pulled out of the inner product:

$$\|x\|^2 = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} c_i \overline{c_j} \langle \phi_i, \phi_j \rangle$$

Applying the orthonormality condition $\langle \phi_i, \phi_j \rangle = \delta_{ij}$ reduces the double sum to a single sum:

$$\|x\|^2 = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \overline{c_i} c_j \delta_{ij}$$

Evaluating the summation over j isolates the non-zero terms where $j = i$:

$$\|x\|^2 = \sum_{i=1}^{\infty} c_i \overline{c_i}$$

Recognizing that $c_i \overline{c_i} = |c_i|^2$, we obtain Parseval's identity:

$$\|x\|^2 = \sum_{i=1}^{\infty} |c_i|^2$$

Completeness ensures that no limit vectors are "missing" from the space, guaranteeing that any physical state represented by an infinite vector expansion has a well-defined norm and remains strictly within the state space of the system.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 12 Assam University questions for this unit in the paper bank**, from PHYDSC-101T 1st Sem 2023 and MATDSC-101T 1st Sem 2023, marked [AU] with their real source. The long-form questions are from other sources, marked as such.

Short-form questions

S1. [AU] Show that vector product of two vectors is a vector. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec A).

Answer in brief: Section 1 — the cross product of two vectors in R^3 is again a vector in R^3, with components given by the determinant form; it is closed under the vector-space operations.

S2. [AU] Define singular and non-singular matrices. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec A).

Answer in brief: Section 4 — a square matrix is singular if its determinant is zero (no inverse exists); non-singular if its determinant is non-zero (inverse exists).

S3. [AU] Show that any square matrix can be uniquely expressed as the sum of symmetric and antisymmetric matrix. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec B, 5 marks).

Answer in brief: Section 7 — write $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$; the first term is symmetric, the second antisymmetric, and the split is unique.

S4. [AU] Show that the set {(1,0,0), (1,1,0), (1,1,1)} is LI. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec A).

Answer in brief: Section 1 — set $a(1,0,0) + b(1,1,0) + c(1,1,1) = (0,0,0)$; solving gives $a = b = c = 0$, so the set is linearly independent.

S5. [AU] Prove that every singleton set containing non-zero vector is LI. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 2 marks).

Answer in brief: Section 1 — for $\{v\}$ with $v \neq 0$, $av = 0$ forces $a = 0$, so the set is linearly independent.

S6. [AU] Show that the vectors (1,1,0), (2,3,5) and (2,2,0) in R^3 are LD. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 3 marks).

Answer in brief: Section 1 — the determinant of the matrix with these vectors as rows (or columns) is zero, so a non-trivial combination gives the zero vector; the set is linearly dependent.

S7. [AU] State and prove Cauchy-Schwarz inequality. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 4 marks).

Answer in brief: Section 2 — $|\langle u, v\rangle| \leq \|u\|\|v\|$, with equality only when u and v are linearly dependent; prove from $\|u - \lambda v\|^2 \geq 0$.

S8. [AU] What do you mean by canonical form of matrices? Define rank of a matrix. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec A).

Answer in brief: Section 1 — the normal (canonical) form is the diagonal form with 1s and 0s obtained by elementary row and column operations; the rank is the number of non-zero rows in that form, i.e. the dimension of the row (or column) space.

S9. [AU] Show that the rank of the transpose of a matrix is the same as that of the original matrix. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 5 marks).

Answer in brief: Section 1 — row rank equals column rank; transposing swaps rows and columns, so the rank is unchanged.

S10. [AU] Solve by Gaussian elimination method: x+y+z=6, 2x-y+2z=6, 2x+2y+z=9. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 5 marks).

Answer in brief: Section 1 — reduce the augmented matrix by row operations to triangular form and back-substitute; the worked example in the notes gives $x = 1, y = 2, z = 3$.

S11. [AU] Find the rank of the matrix [4x5 matrix] by reducing it to normal form. — MATDSC-101T, 1st Sem, 2023, FYUG (Sec B, 5 marks).

Answer in brief: Section 1 — apply elementary row and column operations to reach the normal form; the rank is the number of 1s on the diagonal.

S12. [AU] Find the inverse of the matrix [3x3 matrix]. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec B). Note: the matrix entries are garbled in the paper-bank extraction, so the exact matrix cannot be read from the bank.

Answer in brief: Sections 4 and 6 — by the adjoint method, or by Cayley-Hamilton: A^{-1} follows from the characteristic equation $A^3 - (\operatorname{tr} A)A^2 + \cdots = 0$.

Long-form questions

L1. [non-AU] The eigenvalues of a Hermitian matrix are all — (A) real (B) imaginary (C) of modulus one (D) real and positive. — GATE PH 2018, Q1.

What to write: Section 9 — (A) real; prove it from $A^\dagger = A$ and $Ax = \lambda x$, taking the scalar product with x.

L2. [standard] State and prove the Cayley-Hamilton theorem and use it to find the inverse of a given 3x3 matrix.

What to write: Section 6 in full — every square matrix satisfies its own characteristic equation; then rearrange the equation to express A^{-1} as a polynomial in A.

L3. [standard] Apply the Gram-Schmidt orthogonalization technique to construct an orthonormal basis from a given set of vectors in R^3, and state the Schwarz inequality.

What to write: Sections 2–3 in full — the Schwarz inequality $|\langle u, v\rangle| \leq \|u\|\|v\|$, then the Gram-Schmidt steps: subtract projections, normalize, with the worked example.

L4. [standard] Show that the eigenvalues of a Hermitian matrix are real and that eigenvectors for distinct eigenvalues are orthogonal; define orthogonal and unitary matrices and give examples.

What to write: Sections 7–9 in full — orthogonal: $A^T A = I$; unitary: $A^\dagger A = I$; Hermitian: $A^\dagger = A$; the two proofs from Section 9, with examples (rotation matrix, Pauli matrices).

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.